

ORANGE COUNTY AP, TX, USA

WMO: 723629

Lat: 30.069N

Lon: 93.804W

Elev: 13

StdP: 14.69

Time Zone: -6.00 (NAC)

Period: 05-19

WBAN: 53998

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.9	32.6	18.6	14.1	40.9	23.4	17.6	45.8	18.8	57.4	17.3	56.9	3.0	340	0.365

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	15.6	93.9	78.1	92.0	77.7	90.7	77.5	82.2	87.9	81.2	86.6	80.3	85.6	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
81.0	161.1	84.5	79.8	154.8	83.5	79.0	150.4	82.9	45.8	88.1	44.6	86.2	43.7	85.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
17.1	15.6	13.1	DB	26.0	97.2	3.4	2.3	23.6	98.8	21.6	100.2	19.7	101.5	17.2	103.1
			WB	24.4	82.4	3.7	1.7	21.7	83.6	19.5	84.6	17.4	85.5	14.7	86.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	68.5	50.8	56.1	62.5	68.1	75.2	80.8	82.0	82.8	78.5	69.8	60.7	54.2	(6)
(7)		DBStd	12.96	8.78	9.53	8.21	6.33	5.07	3.04	2.64	2.62	4.19	7.48	8.98	9.20	(7)
(8)		HDD50	229	98	45	11	1	0	0	0	0	0	0	14	60	(8)
(9)		HDD65	1489	445	270	145	43	5	0	0	0	0	39	188	354	(9)
(10)		CDD50	6992	123	217	399	543	781	924	992	1017	855	615	335	191	(10)
(11)		CDD65	2776	5	22	68	135	321	474	527	552	405	189	59	19	(11)
(12)		CDH74	26028	5	40	195	702	2616	5078	5738	6105	3653	1585	261	50	(12)
(13)		CDH80	9804	0	1	8	57	668	2069	2447	2708	1410	418	18	0	(13)
(14)	Wind	WSAvg	5.4	6.1	6.6	6.7	6.7	6.1	5.0	3.8	3.8	3.8	4.6	5.2	5.9	(14)
(15)	Precipitation	PrecAvg	59.0	5.4	3.9	3.8	4.1	4.7	5.6	5.5	4.7	5.5	5.6	4.7	5.2	(15)
(16)		PrecMax	79.3	13.4	8.1	8.0	7.8	11.7	12.4	12.1	10.0	15.0	19.6	10.9	12.2	(16)
(17)		PrecMin	38.4	0.7	0.8	0.7	0.6	0.1	0.3	1.6	0.6	1.1	0.1	0.8	2.5	(17)
(18)		PrecStd	11.1	3.1	2.3	1.8	2.2	3.1	3.1	2.8	2.8	3.3	5.3	2.7	2.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	73.4	78.2	80.9	83.9	90.3	96.1	94.7	97.0	95.3	90.1	82.2	78.2	(19)
(20)			MCWB	65.2	69.4	67.7	68.2	72.2	76.4	76.5	77.1	78.7	76.9	73.4	70.3	(20)
(21)		2%	DB	71.6	74.9	78.4	81.7	87.7	91.4	92.8	94.1	91.7	87.1	79.4	75.2	(21)
(22)			MCWB	64.0	68.0	68.6	69.8	74.2	76.2	77.2	79.1	77.4	74.6	70.7	69.5	(22)
(23)		5%	DB	68.4	72.5	76.6	80.0	85.8	90.3	91.2	92.1	89.9	84.4	77.2	72.5	(23)
(24)			MCWB	63.1	67.4	67.8	69.6	74.2	76.5	77.4	78.8	76.6	73.8	69.1	68.0	(24)
(25)		10%	DB	65.6	70.0	73.5	78.4	83.8	88.9	90.0	90.7	87.8	81.9	74.5	69.9	(25)
(26)			MCWB	61.2	65.9	66.6	69.1	73.9	76.6	77.4	78.5	76.1	71.5	68.7	66.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	69.9	73.0	73.1	76.2	79.5	82.4	82.4	84.2	82.1	80.5	75.6	73.2	(27)
(28)			MCDWB	71.5	75.5	77.3	79.6	84.4	87.0	88.0	90.6	88.0	85.9	79.7	76.1	(28)
(29)		2%	WB	67.4	71.3	71.7	74.2	78.1	80.5	81.2	82.7	80.5	78.8	73.5	71.5	(29)
(30)			MCDWB	69.1	73.5	75.2	78.0	83.0	85.4	86.5	88.9	86.3	83.3	77.1	73.6	(30)
(31)		5%	WB	65.0	69.4	70.5	73.0	77.0	79.6	80.3	81.6	79.6	77.3	72.3	69.8	(31)
(32)			MCDWB	66.8	71.0	73.6	77.0	81.8	85.1	85.6	87.4	85.1	81.4	75.6	71.9	(32)
(33)		10%	WB	62.4	67.2	69.4	71.9	75.8	78.6	79.5	80.6	78.5	75.6	70.4	66.7	(33)
(34)			MCDWB	64.3	68.9	72.1	75.8	80.9	84.5	84.8	86.0	83.5	79.2	73.3	69.0	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	19.1	18.0	18.0	17.7	16.0	15.5	15.5	17.2	20.3	20.3	18.4	(35)	
(36)			MCDBR	20.6	17.2	16.3	17.5	16.9	17.8	17.3	17.6	18.4	19.0	18.5	19.0	(36)
(37)			MCWBR	15.2	11.4	9.1	8.4	6.6	5.9	6.2	5.9	6.8	8.5	11.0	13.7	(37)
(38)		5% WB	MCDWB	18.1	14.5	13.5	13.4	13.2	14.3	15.6	15.2	15.5	16.5	15.2	17.3	(38)
(39)			MCWBR	15.8	11.7	9.2	8.4	6.2	6.2	6.7	6.1	6.4	10.0	10.9	14.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.340	0.357	0.375	0.430	0.471	0.499	0.500	0.483	0.456	0.384	0.371	0.351	(40)	
(41)		taud	2.524	2.484	2.425	2.274	2.190	2.150	2.189	2.239	2.279	2.467	2.444	2.495	(41)	
(42)		Ebn at Noon	284	288	289	275	263	254	253	257	260	275	269	273	(42)	
(43)		Edh at Noon	28	32	36	43	47	49	47	44	41	32	30	28	(43)	
(44)	All-Sky Solar Radiation	RadAvg	908	1028	1430	1725	1932	1965	1892	1792	1582	1387	1047	803	(44)	
(45)		RadStd	88	171	128	124	132	148	153	156	159	177	103	90	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (36 neighbors)	+0.61	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+198	+153

Nomenclature: See separate page